

A POLLON Q OPERATING INSTRUCTION

Thank you for choosing the Apollon sensor from Sentinum. Please read the following operating instructions carefully to prevent damage to the sensor, yourself and the environment.



Table of contents

1. General warnings and safety instructions.....	1
2. Further documentation.....	3
3. Dimensions and components of the sensor	4
3.1. Technical drawing Apollon Q TR / SW (with radar)	4
3.2. Technical drawing Apollon Q T / PP (without radar)	5
4. Scope of delivery and versions.....	6
4.1. Scope of delivery and optional mounting accessories.....	6
4.2. Structure of the product versions	6
4.3. Explanation of the measurement of various sensors	7
4.4. Differences of the iQ versions.....	7
4.5. Availability of the different versions	8
5. Locating and localization methods overview.....	9
5.1. Comparison of different technologies.....	10
5.2. Intelligent use of different tracking technologies for energy optimazation	10
5.3. Which sensors use the Wi-Fi SSID scan?	11
5.4. How does the Wi-Fi SSID scanning work?.....	12
5.5. Advantages and applications of Wi-Fi-based localization.....	12
5.6. Dependence of the accuracy.....	13
5.7. Why tracking via Wi-Fi SSID is usefull for Apollon.....	14
5.8. Functionality of GNSS scans.....	14
5.9. GNSS scan and LoRa® cloud.....	15
5.10. How GPS works	17
5.11. Ultrawideband.....	17
5.12. Tracking in LoRaWAN®	18
5.13. Tracking via the „Cell Locate“ mobile network	20
5.14. eDRX: on its way to the queryable tracker	20
6. Assembly and installation.....	22
6.1. Warnings and safety instructions for mounting	22
6.2. Recommended fastening methods.....	23
6.3. Mounting accessories	23
6.4. General installation instructions	24
6.5. Important note for devices with an external antenna	24
6.6. Wall mounting with screws.....	25
6.7. Wall mounting with magnets	25
6.8. Wall mounting with adhesive strips	26

6.9. General Information on installation with blind rivet nuts.....	26
6.10. Front panel installation with rivet nuts	29
6.11. Rear panel installation with blind rivets	30
6.12. Front panel installation with blind rivets.....	31
7. Commissioning and use	32
7.1. Commissioning the sensor via BLE	32
7.2. NFC commissioning, parameter configuration and NFC Tag location	35
7.3. Possible applications of the sensors	36
8. Flap opening detection and tilt detection	37
8.1. Flap opening detection via magnetic switch	37
8.2. Flap opening detection via accelerometer and tilt detection	37
8.3. Orientations (example Apollon IQ).....	38
9. Communication with the interface	40
9.1. LoRaWAN join behavior	40
9.2. Mioty join behavior	40
9.3. Cellular join behavior (NB-IoT and LTE-M1).....	41
10. Maintenance and cleaning.....	43
11. Batterie change	44
12. Marking and certification	45
13. Drill templates for installation.....	46
13.1. Blind rivet nut mounting.....	46
13.2. Blind rivet mounting	47
13.3. Wall mounting (see next page).....	47

1. GENERAL WARNINGS AND SAFETY INSTRUCTIONS

Warnings and important information about potential hazards or damage

Important instructions necessary for the smooth operation of the devices

Please note:

- Observe the safety instructions and installation instructions in the manual and the installation list.
- Ensure that the installation environment complies with the prescribed application area guidelines. Observe the temperature and other limit values at all times.
- The device may only be used in the areas specified in the technical specifications.
- The device may only be used for the purposes and in the areas described.
- Safety and functionality can no longer be guaranteed if the device is modified or extended.
- The sensor must not be mounted on ceilings or floors
- Operation of the sensor is only permitted up to a maximum of 2000 meters above sea level.
- Due to human exposure regulations, a minimum distance of 20 cm must be maintained between the device and persons.
- Ensure that the installation environment complies with the prescribed guidelines for the area of application. Observe the temperature and other limit values at all times.
- Ensure that the installation environment complies with the prescribed guidelines for the area of use. Comply with the temperature and other limit values at all times.
- Dispose of the appliance in accordance with national environmental regulations.
- The device may only be repaired by the manufacturer.
- Observe the specific safety instructions for batteries and lithium primary cells.

If the device is installed incorrectly:

- It may not function properly.
- It could be permanently damaged.
- It could pose a risk of injury.

Please note:

- Improper handling, such as improper mechanical stress, e.g. dropping the device, may result in damage.
- If battery cells other than those recommended are used, performance and product safety may be adversely affected.

2. FURTHER DOCUMENTATION

Please note the information and limit values in the [technical data sheet](#).

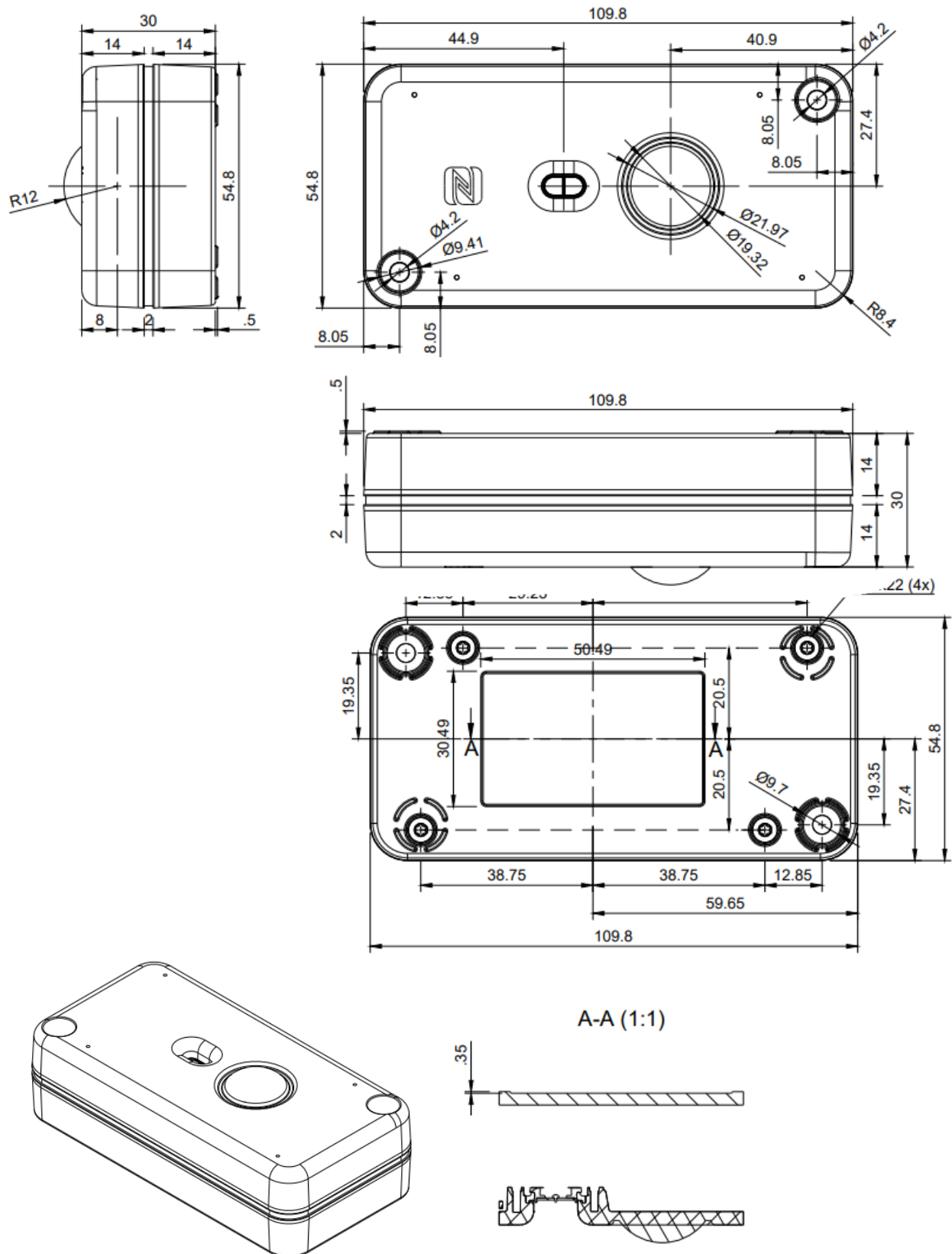
The sensor-specific factory settings (Sentiface), as well as the keys and permissible values of the sensor can be found in the [NFC and downlink description](#). The Senticom and Sentivisor tables can be found in the [generic NFC and downlink documentation](#). The special functions for [vandalism detection and opening detection](#) are also generic.

The option to configure sensor communication can be found in the respective generic [LoRaWAN®](#), [Mioty®](#) or [Cellular \(NB-IoT and LTE-M1\)](#) documentation, depending on the version.

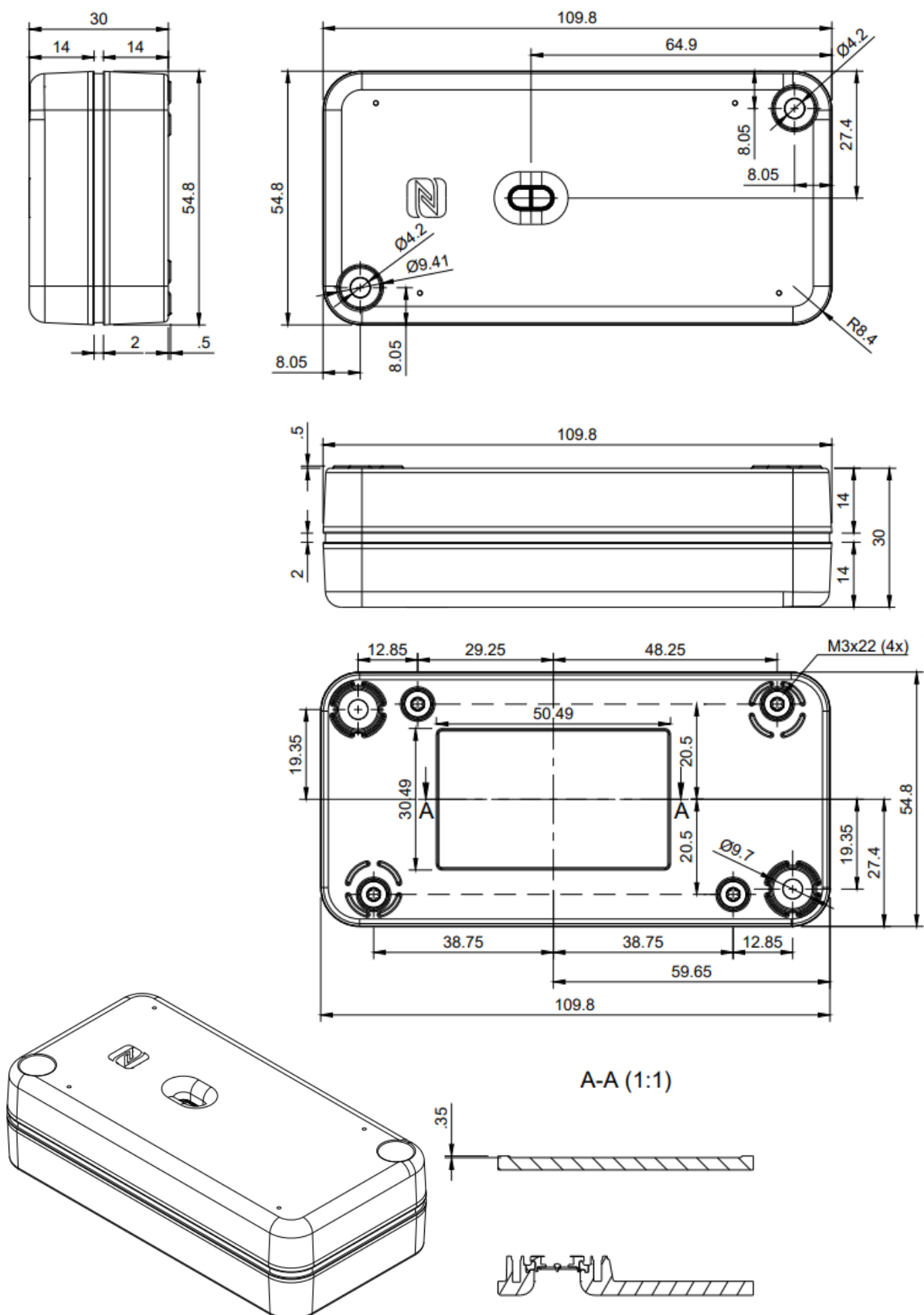
All documents for the generic documentation can be found at <https://docs.sentinum.de/en/important-cross-product-documentation-for-sensors>.

3. DIMENSIONS AND COMPONENTS OF THE SENSOR

3.1. TECHNICAL DRAWING APOLLON Q TR / SW (WITH RADAR)



3.2. TECHNICAL DRAWING APOLLON Q T / PP (WITHOUT RADAR)



4. SCOPE OF DELIVERY AND VERSIONS

4.1. SCOPE OF DELIVERY AND OPTIONAL MOUNTING ACCESSORIES

Art. number	Recommended accessories
Z-APOQ-MAG-NEO	Magnet set APOQ ferrite
Z-APOQ-MAG-FER	Magnet set APOQ ferrite
Z-APOQ-RWM-BN	Rear panel mounting set with blind rivets
Z-APOQ-RWM-NM	Rear panel mounting set with rivet nuts M4
Z-APOQ-VWM-BN	Front panel mounting set with blind rivets
Z-APOQ-VWM-NM	Front panel mounting set with rivet nuts M4
Z-APOQ-AP	Spacer plate PA
Z-APOQ-KLEBER	Mounting adhesives
Z-APOQ-SCHRAUBEN	Machine screws DIN912 with nuts M4
Z-APOQ-SPX	Wood screws

Drilling templates can be found at the end of the operating instructions

The scope of delivery includes:

- 2x batteries (already inserted)
 - For LoRaWAN® and mioty® sensors
 - Energizer® Ultimate Lithium-Batterien - AA
 - VARTA ULTRA LITHIUM Mignon AA
 - For cellular sensors (NB-IoT and LTE-CAT-M1)
 - VARTA-CR-AH-R
- Apollon Sensor in one of the following versions

4.2. STRUCTURE OF THE PRODUCT VERSIONS

PREFIX-PRODUCT-SERIES-COMMUNICATION-SENSORSTECHNOLOGY-
OTHERSENORTECHNOLOGY

Prefix	Product range	Communication	Sensor technology	Other sensor technology
S	APOQ (APOLLON-Q)	mioty	T (ToF)	ACC (accelometer = acceleration sensor for detecting occurring movements)
	iAPOQ (industrial version)	LoEU (LoRaWAN®)	R (radar, fill level)	ACC-HALL (accelometer = acceleration sensor + hall = Hall sensor)
		NBM1 (LTE-CAT-M1)	TR (ToF and radar)	
		NB (NB-IoT)	SW (Smart Waste, TR and other software)	
			PP*	

* Not in combination with additional sensors!

4.3. EXPLANATION OF THE MEASUREMENT OF VARIOUS SENSORS

Name	Explanation
LOEU	LoRaWAN EU
MIOTY	mioty EU
NBM1	NB-IoT, LTE-CAT-M1
SW	Optimized radar for Smart Waste, especially for underfloor containers and heavily soiled areas
T	ToF
R	Radar
TR	Dual measuring principle of ToF and radar
ACC	Acceleration sensor for flap opening and vandalism detection
TR (without SW)	For liquids and good reflectors
SMAV	Small Angle of View 18°

4.4. DIFFERENCES OF THE IQ VERSIONS

The Apollon-iQ versions have extended certifications and security functions for industrial use. These include CRA conformity with integrated secure elements in hardware and software.

The iQ models also offer an extended range of functions and additional configuration options, particularly with regard to measurement and transmission intervals. For example, significantly shorter intervals can be implemented than with the standard series, which enables more precise and dynamic measurement applications. Firmware updates can also be carried out on the iQ versions via FOTA (Firmware Over-the-Air), which makes maintenance in the field considerably easier.

The Apollon iQ series also stands out visually with its yellow seals and is particularly suitable for critical and long-term stable industrial applications thanks to comprehensive end-of-line tests and the use of the latest components.

4.5. AVAILABILITY OF THE DIFFERENT VERSIONS

sensors name	GNSS	opening detektion	available	configuration
S-APOQ-LOEU-T	X	X	✓	NFC
S-APOQ-MIOTY-T	X	X	✓	NFC
S-APOQ-LOEU-T-ACC	✓	✓	✓	NFC/BLE
S-APOQ-LOEU-TR-ACC	✓	✓	✓	NFC/BLE
S-APOQ-LOEU-SW-TR-ACC	✓	✓	✓	NFC/BLE
S-APOQ-MIOTY-T-ACC	X	✓	✓	NFC/BLE
S-APOQ-MIOTY-TR-ACC	X	✓	✓	NFC/BLE
S-APOQ-MIOTY-SW-TR-ACC	X	✓	✓	NFC/BLE
S-APOQ-NBM1-T-ACC	✓	✓	✓	NFC/BLE
S-APOQ- NBM1-TR-ACC	✓	✓	✓	NFC/BLE
S-APOQ- NBM1-SW-TR-ACC	✓	✓	✓	NFC/BLE
S-APOQ-LOEU-R-ACC	✓	✓	✓	NFC/BLE
S-APOQ-MIOTY-R-ACC	X	✓	✓	NFC/BLE
S-APOQ-NBM1-R-ACC	✓	✓	✓	NFC/BLE
S-APOQ-LOEU-T-SMAV	X	X	✓	NFC/BLE
S-APOQ-MIOTY-T-SMAV	X	X	✓	NFC/BLE
S-APOQ-LOEU-T-ACC-SMAV	✓	✓	X	NFC/BLE
S-APOQ-LOEU-TR-ACC-SMAV	✓	✓	X	NFC/BLE
S-APOQ-LOEU-SW-TR-ACC-SMAV	✓	✓	X	NFC/BLE
S-APOQ-MIOTY-T-ACC-SMAV	X	✓	X	NFC/BLE
S-APOQ-MIOTY-TR-ACC-SMAV	X	✓	X	NFC/BLE
S-APOQ-MIOTY-SW-TR-ACC-SMAV	X	✓	X	NFC/BLE
S-APOQ-NBM1-T-ACC-SMAV	✓	✓	X	NFC/BLE
S-APOQ- NBM1-TR-ACC-SMAV	✓	✓	X	NFC/BLE
S-APOQ- NBM1-SW-TR-ACC-SMAV	✓	✓	X	NFC/BLE

* iAPOQ versions on request

5. LOCATING AND LOCALIZATION METHODS OVERVIEW

In general, all LoRaWAN® devices can be located via the LoRaWAN®. This requires gateways with GPS synchronization.

As of Q2 2025, the Apollon-Q and Juno series in the Sentinum product portfolio are equipped with extended tracking functions.

Tracking via BLE function can be activated on request.

sensor number	communication standard	Wi-Fi SSID scan	GNSS scan	GNSS	Cell locate
S-JUNO(-iX)-LOEU-TRACK	LoRaWAN®	✓	✓	X	X
S-JUNO(-iX)-LOEU-TH-TRACK	LoRaWAN®	✓	✓	X	X
S-JUNO(-iX)-NBM1-TRACK-2	Cellular	✓	X	✓	✓
S-JUNO(-iX)-NBM1-TRACK-3	Cellular	✓	X	✓	✓
S-JUNO(-iX)-NBM1-TH-TRACK-2	Cellular	✓	X	✓	✓
S-JUNO(-iX)-NBM1-TH-TRACK-3	Cellular	✓	X	✓	✓
S-JUNO(-iX)-MIOTY-TRACK	mioty®	✓	X	✓	X
S-JUNO(-iX)-MIOTY-TH-TRACK	mioty®	✓	X	✓	X
S-(i)APOQ-LOEU-T-ACC	LoRaWAN®	✓	✓	X	X
S-(i)APOQ-LOEU-TR-ACC	LoRaWAN®	✓	✓	X	X
S-(i)APOQ-NBM1-T-ACC	Cellular	✓	X	✓	✓
S-(i)APOQ-NBM1-TR-ACC	Cellular	✓	X	✓	✓

In the case of cellular sensors, TRACK 2 versions differ from TRACK 3 versions in the number of battery cells to be used; the number 2 or 3 describes these. In addition, SIM cards can be used if only two cells are used.

5.1. COMPARISON OF DIFFERENT TECHNOLOGIES

technology	range under good conditions (m)*	Range under bad conditions (m)*	power consumption	suitability for indoor tracking	costs
BLE scanning (not implemented, on request)	1 - 3	5 - 10	Low	High	Medium
Wi-Fi SSID scanning	1 - 5	5 - 20	Low	High	Medium
GNSS (GPS, Glonass, BeiDou, Galileo)	3 - 5	5 - 10	High	Not suitable	High
mobile radio localization via triangulation or radio cells	10 - 150	150 bis mehrere Kilometer	Low	Low	Low
GNSS scanning	1 - 10	10 - 200	Low	Not suitable	Medium Low
UWB	<0,1 - 0,3	0,3 - 0,5	Low	Very high	Medium
tracking via LoRaWAN***	200 - 500	500 - 1500	None	Practically not suitable	Low

*good conditions: few shadowing effects, direct connections to satellites or gateways, very good antennas

**bad conditions: large shadowing effects such as trees, buildings or walls, poor connection quality

***depends on the number of gateways and gateway equipment

5.2. INTELLIGENT USE OF DIFFERENT TRACKING TECHNOLOGIES FOR ENERGY OPTIMAZATION

We combine various tracking technologies for precise and energy-efficient location determination: Wi-Fi SSID scan, GNSS, GNSS scan and Cell Locate. Each of these technologies has specific strengths that we use flexibly and depending on the situation.

- Wi-Fi SSID scan detects available Wi-Fi networks in the vicinity and determines a position based on known SSID locations. This method is extremely energy-efficient and enables fast location updates - ideal in urban areas with dense Wi-Fi coverage.
- GNSS (Global Navigation Satellite System, e.g. GPS) offers very precise positioning, but is very energy-intensive in comparison. GNSS is therefore only activated when other methods do not provide sufficiently accurate data.
- With GNSS scanning, satellite data is only collected and the position calculation is optimized without permanently maintaining an active GNSS session. This also saves considerable energy compared to permanent GNSS use.

- Cell Locate enables positioning based on mobile radio cells. This method is available globally and provides a rough but continuous location determination even if there is no WLAN or GNSS signal.

Intelligent control and prioritization - such as the preferred use of Wi-Fi scans - can significantly reduce the device's energy consumption. Only when Wi-Fi or cell positioning is not sufficient does the device automatically switch to GNSS or other more precise methods.

All tracking strategies and fallback mechanisms are individually configurable so that the best balance between energy efficiency, accuracy and availability can be selected for different applications and regions.

Even if individual technologies vary depending on the environment, this flexible combination enables virtually seamless and detailed route recording worldwide.

We also offer suitable solutions for indoor scenarios where GNSS signals are often unavailable or inaccurate. Thanks to the Wi-Fi SSID scan and the use of known indoor access points, the position can also be reliably determined inside buildings. Optionally, indoor positioning can be supplemented by additional technologies such as Bluetooth Low Energy (BLE) beacons or inertial sensors.

This enables precise location determination even in complex environments such as shopping centers, airports or industrial halls - seamlessly integrated into the existing tracking concept.

5.3. WHICH SENSORS USE THE WI-FI SSID SCAN?

In the Sentinum product portfolio, parts of the Juno and Apollon-Q series are equipped with a Wi-Fi SSID scan function. The Cellular and mioty® products are equipped with a 2.4 GHz and 5 GHz Wi-Fi SSID scan for the evaluation of up to 20 MAC addresses, the LoRaWAN® products with a 2.4 GHz Wi-Fi SSID scan for up to six addresses:

article number	communication standard	2,4 GHz scan	5 GHz scan	Maximum number of MAC addresses
S-JUNO(-iX)-LOEU-TRACK	LoRaWAN®	✓	X	6
S-JUNO(-iX)-LOEU-TH-TRACK	LoRaWAN®	✓	X	6
S-JUNO(-iX)-NBM1-TRACK-2	Cellular	✓	✓	20
S-JUNO(-iX)-NBM1-TRACK-3	Cellular	✓	✓	20
S-JUNO(-iX)-NBM1-TH-TRACK-2	Cellular	✓	✓	20
S-JUNO(-iX)-NBM1-TH-TRACK-3	Cellular	✓	✓	20
S-JUNO(-iX)-MIOTY-TRACK	mioty®	✓	✓	20
S-JUNO(-iX)-MIOTY-TH-TRACK	mioty®	✓	✓	20
S-(i)APOQ-LOEU-T-ACC	LoRaWAN®	✓	X	6
S-(i)APOQ-LOEU-TR-ACC	LoRaWAN®	✓	X	6
S-(i)APOQ-NBM1-T-ACC	Cellular	✓	✓	20
S-(i)APOQ-NBM1-TR-ACC	Cellular	✓	✓	20

5.4. HOW DOES THE WI-FI SSID SCANNING WORK?

Wi-Fi SSID scan-based localization uses the detection of Wi-Fi networks in the environment to determine the location of a device. It uses the signal strength (RSSI) of the Wi-Fi signals to estimate the distance to the various access points (APs). This technique is often used indoors where GPS signals may be unavailable or inaccurate.

Essentially, the localization process works as follows:

1. activation of the SSID scan: The device starts with a passive Wi-Fi scan, during which it listens for all beacon frames emitted by the access points (APs) in the vicinity. These beacon frames contain the SSID (the name of the network), the BSSID (the MAC address of the AP) and the signal strength of the received signal (RSSI).
2. measurement of the signal strength: The RSSI (Received Signal Strength Indicator) is measured for each signal received. This signal strength indicates how strongly the signal from the access point is received by the device. A higher RSSI usually means that the device is closer to the corresponding access point.
3. comparison with known positions: To determine the position of the device, the measured RSSI value is used in combination with the known positions of the access points. This is done using methods such as triangulation or trilateration, in which the distances to at least three or more access points are calculated. Using these calculations, the device can determine its position on a map or in a room.
4. positioning: The data collected from the access points is analyzed to determine the most likely position of the device. This is done taking into account the signal strength and the known positions of the access points. Modern algorithms can also further refine the localization by taking into account additional factors such as environmental conditions or the movements of the device.
5. display of the position: Once the device has determined the position, this is displayed to the user on a map or in a corresponding user interface. If required, the accuracy of the position can also be updated in real time, based on further SSID scans and changing signal strengths.

5.5. ADVANTAGES AND APPLICATIONS OF WI-FI-BASED LOCALIZATION

- High accuracy indoors: As GPS signals are often weak or non-existent indoors, Wi-Fi scanning offers an excellent alternative to indoor positioning.
- Easy to implement: As many buildings are already equipped with Wi-Fi networks, localization via Wi-Fi can be implemented with minimal additional effort.
- Cost-effective: Wi-Fi-based localization requires no additional hardware investment if Wi-Fi access points are already in place.

This localization approach is particularly advantageous in indoor navigation systems, asset tracking or fleet management, as it enables precise positioning, even without the use of expensive GPS systems.

Positioning technology based on Wi-Fi SSID scanning is used in a variety of applications where rough to medium positioning accuracy is required and existing Wi-Fi infrastructure can be used. A device records the names (SSIDs) and signal strengths (RSSI) of the surrounding Wi-Fi networks without connecting to them. This information can be used to

estimate where the device is located - either by comparing it with an existing Wi-Fi database (e.g. from Google or Apple), by prior fingerprinting or with the help of a self-built Wi-Fi mapping model.

This technology is used, for example, for indoor localization in buildings, such as shopping malls, airports or large office complexes, where GPS signals are only available to a limited extent or not at all. Wi-Fi scanning is also used in logistics and asset tracking to track the location of devices or goods within warehouses or on company premises - often in combination with other technologies such as BLE or LoRaWAN. In smartphones and wearables, the method is used to enable location-based services such as map navigation, geofencing or location sharing, even when no mobile connection is available.

Another typical use case is assisted positioning for battery-powered IoT devices where GNSS would be too energy-intensive. Here, Wi-Fi Scan can help to determine a sufficiently accurate position without putting a heavy strain on the battery. Due to the wide availability of Wi-Fi networks and the ability to estimate a position even without an active network connection, Wi-Fi SSID scanning is a flexible, cost-effective and energy-saving alternative or supplement to classic GNSS or cellular positioning systems.

5.6. DEPENDENCE OF THE ACCURACY

The accuracy of Wi-Fi SSID scan localization depends on several factors and can vary greatly depending on the environment and system architecture. In general, the accuracy of position-based localization using Wi-Fi is typically between 1 and 20 meters. Exact positioning depends on various parameters that are both technical and environmental in nature.

A key influencing factor is the signal strength, also known as RSSI (Received Signal Strength Indicator). This is used to estimate the distance between the device and an access point. However, the relationship between signal strength and distance is not linear and can be significantly distorted by walls, furniture, other devices or structural conditions. In open spaces with few obstacles, the accuracy can be around 1 to 5 meters, while in more complex environments such as residential or office buildings with many obstacles, it tends to fluctuate between 5 and 20 meters.

The number and distribution of available access points also plays a decisive role. The more access points with a known and stable position are available, the more precise the positioning can be. If the device receives signals from at least three well-distributed access points, more precise positioning is possible through triangulation or trilateration. If, on the other hand, only one or two access points are available or if they are unfavorably distributed, the accuracy decreases accordingly.

The environment also has a significant influence on localization accuracy. Wi-Fi signals can be reflected, absorbed or scattered - depending on the material and arrangement of walls, furniture or other objects in the room. Nearby electronic devices can also cause interference. In a densely built-up office building with many steel beams, partition walls and other obstacles, for example, the localization accuracy can be between 5 and 15 meters, while in an open storage room or corridor it can be between 1 and 3 meters.

In addition to the hardware, the software also plays an important role: the quality of the positioning algorithms used can significantly improve accuracy. Methods based on machine learning or calibration data, for example, can take better account of environmental influences and help to interpret and smooth the measured RSSI values. This increases the reliability of the position information, especially in more complex environments.

Another aspect is the visibility of the networks. Hidden SSIDs, i.e. WLANs that do not broadcast their network name, as well as interference from neighboring networks or other devices, can also reduce positioning accuracy. In such cases, the device has less usable information at its disposal, which can lead to less accurate positioning.

Overall, Wi-Fi scanning is a flexible and, in many cases, sufficiently accurate method for determining position - especially when other positioning technologies such as GPS are not available or are too energy-intensive. However, the accuracy that can be achieved always depends on the interaction of various factors.

5.7. WHY TRACKING VIA WI-FI SSID IS USEFULL FOR APOLLON

The Juno with Wi-Fi SSID Scanning is particularly suitable for positioning applications where GNSS signals are weak or unavailable, such as indoors or in urban environments. By combining Wi-Fi scanning with GNSS and cell-based positioning technologies, more precise location data can be achieved.

- Wi-Fi scanning: supports both active and passive scanning of 2.4 GHz and 5 GHz Wi-Fi networks.
- energy efficiency: optimized for low power consumption applications, ideal for battery-powered devices.
- high security with WPA3 support
- very good indoor detection method
- many existing networks
- can significantly extend service life as much more power efficient than conventional GNSS

Apollon Cellular can evaluate up to 20 different access points and thus provide very precise localization. The accuracy varies depending on the application. Accuracies of 3 to 20 meters are realistic.

5.8. FUNCTIONALITY OF GNSS SCANS

The GNSS scan functionality makes it possible to determine position data via various GNSS systems such as GPS, Galileo, GLONASS or BeiDou. The position is determined by an integrated GNSS receiver, which continuously scans for visible satellites and calculates the position of the device based on the signals received. As soon as the position has been determined, the corresponding data is transmitted to a central station or cloud platform. This enables precise localization, even in remote areas, with low energy consumption as the GNSS receiver is only activated when needed.

The accuracy of the GNSS scan function in LoRa modules depends on various factors. The typical accuracies of GNSS positioning are between 2.5 and 10 meters, depending on the system used, when standard GPS is used. When multiple GNSS systems such as GPS, Galileo, GLONASS and BeiDou are combined, accuracy can be improved to 1 to 3 meters. However, in difficult environments, such as urban mountain canyons or strong signal shadowing, the accuracy can increase to 10 to 50 meters or more.

- Signal reception: The GNSS receiver (e.g. in the smartphone) receives radio signals from at least four GNSS satellites (e.g. GPS, Galileo, GLONASS).
- Time of flight measurement: Each signal contains a time stamp. The receiver measures how long it took for the signal to travel from the satellite to earth (transit time).
- Distance estimation: The distance to each satellite is calculated from the signal transit time (distance = speed of light × transit time).
- Position calculation (trilateration): Using the distances to at least four satellites, the receiver can calculate its own location (longitude, latitude, altitude) and the exact time by calculating the intersection points of the spheres around the satellites.
- Corrections: Errors due to the atmosphere, satellite orbits or clocks are partially corrected by algorithms or additional systems (such as DGPS or SBAS).

5.9. GNSS SCAN AND LORA® CLOUD

The LoRaWAN® devices such as the Juno Tracker or the Apollon-Q are based on the LoRa® Edge LR1110 chipset and send the GNSS scan information on port 199. The data is sent to databases such as the LoRa® Cloud, where the latitude and longitude coordinates are calculated. The coordinates are then returned to the network server via standardized interfaces.

Sentinum offers such a service. You can simply send the data to our servers and we will do the rest for you. Just ask us.

If you want your own integration, the following links will help you:

Connecting TTI to the LoRa Cloud: [LoRa Cloud | The Things Stack for LoRaWAN®](#)

Connecting Chirpstack to the LoRa Cloud: [LoRa Cloud - ChirpStack open-source LoRaWAN® Network Server documentation](#)

LoRa Cloud Homepage: [Semtech LoRa Cloud](#)

Example of the TTI integration:

To connect The Things Stack (TTI) - i.e. the LoRaWAN® platform from The Things Industries - with the Semtech LoRa Cloud, you need to set up an integration so that data, e.g. GNSS scans, are correctly transmitted to the LoRa® Cloud and processed. The LoRa® Cloud takes over tasks such as geolocalization, GNSS conversion, Wi-Fi positioning or modem services.

Requirements:

- an active account with The Things Stack (TTI).
- a registered LoRaWAN device (e.g. a tracker with GNSS).
- API access to Semtech LoRa Cloud Services (via Dev Portal: <https://lora-developers.semtech.com>).
- LoRa Cloud Token (API Key) - you can get this in the LoRa Cloud Portal.

1. activate Semtech LoRa Cloud

- go to <https://lora-developers.semtech.com>.
- create an account or log in.
- under LoRa Cloud → Modem Services you will find your token (API key), which you must enter later in TTI.

2. Set up integration in The Things Stack

- Log in to The Things Stack Console (e.g. <https://eu1.cloud.thethingsindustries/>).
- Open your end device that you want to connect.
- Go to Integrations → Webhooks.
- Click on Add Webhook and select Semtech LoRa Cloud as the template.

3. Configure webhook

- Fill out the form:
 - Base URL: Is automatically suggested by TTI.
 - Token: Enter your API key from the LoRa Cloud here.
 - Activate the desired services, e.g:
 - Modem Services (for GNSS and Wi-Fi scans).
 - Geolocation (for TDOA/RSSI).
 - You can also have GNSS or Wi-Fi data sent, depending on the device type.

4. adapt payload formats (if necessary)

- Make sure that your end device uses the expected payload structure for Semtech LoRa Cloud Services (e.g. the format provided by Semtech's LoRa Basics Modem).

5. check data

- As soon as your device sends position data (e.g. GNSS raw data), it is forwarded to the LoRa Cloud via TTI.
- The response from the LoRa Cloud is then sent back to your end device or application via TTI.

Test & Monitoring:

- use the Live Data view in TTI to see whether data is being transmitted.
- in the Semtech Cloud, you can see whether requests are arriving and being processed.
- check the response packets with the geodata (latitude, longitude) and position accuracy.

Note:

This integration works particularly well with devices based on Semtech's LoRa Basics Modem-E architecture (e.g. with LoRa Edge™ chips such as LR1110), but custom formats are also possible as long as the API requests are compatible.

5.10. HOW GPS WORKS

GPS (Global Positioning System) is a satellite-based navigation system that makes it possible to determine the exact position on earth. It consists of at least 24 satellites orbiting the earth at an altitude of around 20,000 kilometers. These satellites continuously send out signals containing information about their current position and the exact time at which the signal was sent. Each satellite is equipped with an atomic clock that is extremely precise.

A GPS receiver, which is installed in devices such as smartphones, navigation systems or other GPS-enabled devices, receives these signals. To determine its own position, the receiver needs signals from at least four satellites. As soon as the signals are received, the receiver measures the time it took for the signal to travel from the satellite to the receiver. As the light propagates at a constant speed, the receiver can calculate the distance to each satellite.

With the distances to at least three satellites, the receiver can determine the position on the earth's surface using triangulation. A fourth satellite helps to calculate the altitude (the Z coordinate) and correct any errors. The accuracy of the GPS position depends on the number of satellites received and the quality of the signal. The accuracy is best in open areas without obstacles such as tall buildings or trees. In urban mountain canyons or in bad weather, signal interference can affect the accuracy.

To further improve accuracy, differential GPS (DGPS) is used in many cases. Stations are installed at fixed, known points on the earth which send correction data to the mobile GPS receivers in order to increase the accuracy to a few centimetres.

In addition to the American GPS system, there are also other global navigation satellite systems (GNSS) such as GLONASS (Russia), Galileo (Europe) and BeiDou (China), which are often used in conjunction with GPS to improve the accuracy and availability of positioning.

5.11. ULTRAWIDEBAND

Ultra-wideband (UWB) is a modern radio technology for high-precision positioning, which is used particularly indoors. Positioning is achieved by transmitting extremely short and broadband radio signals in the frequency range from around 3.1 to 10.6 GHz. These signals have a very high time resolution, which means that the transit time of the signal - i.e. the time it takes to travel from a transmitter to a receiver - can be measured extremely accurately. Based on this time measurement, the distance between two devices can be calculated with an accuracy of typically 10 to 30 centimeters, in some cases even less than 10 centimeters.

UWB positioning works using two main methods: two-way ranging (TWR) and time difference of arrival (TDoA). In two-way ranging, a mobile device (also known as a tag) sends a radio signal to a permanently installed receiver (anchor). This responds and the tag measures the time required for the outward and return journey. The distance can be calculated from this time, taking into account the constant propagation speed of the radio waves. The TDoA method works slightly differently: here, the tag only transmits a signal

that is received by several anchors simultaneously. The minimum time difference with which the signal arrives at the different receivers is used for triangulated calculation of the position. This method enables particularly energy-efficient applications, as the tag does not have to actively respond and the computing work is carried out on the server side.

A typical UWB positioning system consists of several permanently installed anchors with known positions and mobile tags that are attached to objects, people or vehicles. The position is calculated by a central positioning software that continuously processes the signals. UWB is characterized not only by its high accuracy, but also by its low latency, which makes it ideal for real-time applications - for example in industrial production, in logistics centers or for access control in buildings. Even in complex environments with a lot of metal or other radio sources, UWB remains very reliable due to its high robustness against interference. The typical range indoors is between 30 and 100 meters, depending on the antenna configuration and the structural conditions. This makes UWB an extremely powerful solution for precise, secure and energy-efficient indoor tracking.

5.12. TRACKING IN LORAWAN®

Tracking in the LoRaWAN® network works by end devices (so-called nodes) transmitting radio signals that are received by several LoRaWAN® gateways. The exact position of the device is not determined directly by the device itself, but by evaluating the signals received in the network or in a special positioning platform (e.g. the Semtech LoRa® Cloud). There are various methods for determining position, which can be combined depending on the application and infrastructure.

One frequently used approach is the so-called TDOA method (Time Difference of Arrival). Here, the network measures the time difference with which a radio signal from a LoRaWAN® device arrives at various gateways. As radio signals propagate at the speed of light, these minimal time differences can be used to derive distance differences to the gateways. If at least three gateways receive the same signal, the position of the device can be calculated by triangulation. This calculation is performed centrally in the LoRaWAN® Network Server or in a connected cloud solution. The accuracy of TDOA is usually in the range of around 200 to 1000 meters, depending on gateway density, synchronization and environmental conditions.

The TDOA (Time Difference of Arrival) method in the LoRaWAN® network works on a principle similar to triangulation, or more precisely, it is a variant of multilateration. The position of a device is not calculated directly from the signal strengths (as with the RSSI method), but from the time differences of a radio signal at several gateways.

When a LoRaWAN® end device (node) transmits a message, this signal is received simultaneously (or almost simultaneously) by several gateways within range. Each of these gateways notes with extremely high time resolution exactly when the signal arrived. As the radio signal propagates at the speed of light, even differences in the nanosecond range make a measurable difference to the calculated distance.

By calculating the time differences with which the signal arrives at the various gateways, the system can calculate circles (or hyperbolas) with possible positions of the device. The more gateways receive the signal, the more accurately the intersection of these hyperbolas can be determined - i.e. the actual position of the device. This method requires at least three synchronized gateways to calculate a two-dimensional position (latitude/longitude).

Are special gateways required?

A special type of gateway is required for tracking in the LoRaWAN® via TDOA: These must be GPS-synchronized or have another precise time synchronization (e.g. PTP - Precision Time Protocol) so that the time stamps for receiving the signal are exact and comparable.

Standard LoRaWAN® gateways without time synchronization cannot provide reliable TDOA data, as even the smallest deviations in time recording would lead to major errors in position determination.

Accuracy

The accuracy of TDOA depends heavily on the density and distribution of the gateways, the quality of the time synchronization and the environment (e.g. reflections). Typically, it is in the range of 200 to 1000 meters, in ideal conditions even better. In urban environments, it can be affected by multipath effects (reflections).

TDOA for indoor tracking:

Indoor tracking with the TDOA method in the LoRaWAN® network is theoretically possible, but in practice it is very limited and usually not recommended when it comes to precise localization inside buildings. Here are the reasons in detail:

Why TDOA is problematic for indoor applications:

1. radio wave distortion due to obstacles
walls, ceilings, furniture and other objects cause strong attenuation, scattering and reflections of radio signals. This changes the effective propagation time of the signal, which leads to massive accuracy errors in a method based on time differences such as TDOA.
2. multipath propagation
Radio signals not only reach the gateways directly, but often also via reflections. These signals arrive with minimal delay and distort the time measurement, making the position calculation inaccurate.
3. difficult gateway placement
For meaningful TDOA tracking, there must be at least three gateways with a clear line of sight to the device - this is difficult to achieve in buildings. Even large buildings are often only covered by one gateway, which does not allow TDOA positioning.
4. synchronization suffers from poor GPS reception
GPS synchronization of gateways is often not possible or unreliable indoors, which ruins the basis for TDOA. Without accurate time synchronization, the entire method does not work.

When does TDOA work indoors *with restrictions*:

- In very large halls, airport terminals or open logistics areas with good gateway coverage.
- When additional error correction technologies are used (e.g. algorithms that detect multipath effects).

- In combination with other localization technologies such as Bluetooth, UWB or Wi-Fi to compensate for failures or inaccuracies.

5.13. TRACKING VIA THE „CELL LOCATE“ MOBILE NETWORK

Mobile phone positioning works by a mobile device communicating with the mobile phone network via its radio signal, and the network then calculates the approximate location based on various parameters. Here is an overview of the most important methods:

Method	Typical accuracy	Remark
Cell-ID	100 m - multiple km	Very rough; depends on cell size (city vs. country)
Enhanced Cell-ID	50 - 500 m	Better due to timing info, but depends on network
TDOA (Time Difference of Arrival)	50 - 150 m	Requires several synchronized stations
AOA (Angle of Arrival)	100 - 200 m	Less widespread, requires special antennas

The accuracy of positioning via mobile radio depends heavily on the method used, the network coverage and the environment. The power consumption is very low compared to other technologies.

5.14. EDRX: ON ITS WAY TO THE QUERYABLE TRACKER

The eDRX function is only available for mobile devices!

The dream of a tracker that is constantly listening and can be actively interrogated is becoming a reality. The eDRX function helps to make this a reality.

eDRX (Extended Discontinuous Reception) allows a **mobile device** to switch to an energy-saving "sleep mode" after a data transmission, in which it does not constantly communicate with the mobile network. Normally, mobile devices have to check at short intervals whether the network has new messages for them (e.g. incoming commands or updates). These frequent checks cost energy, even if there is no new data.

With eDRX, these check intervals are significantly extended: a sensor can be set so that it only listens for new network messages every minute or even hour. During idle periods, the device's receiver largely switches off, which drastically reduces energy consumption. As soon as the specified eDRX phase ends, the sensor "wakes up", listens briefly for new messages and can then go back to sleep if nothing important has been received. The location of the device can be queried by sending a message from the network to the sensor.

The device remains registered with the network - it is not completely offline - but only reduces its active readiness to receive messages. This is ideal for applications in which the device mainly sends data itself (e.g. location, sensor readings) and only rarely needs to be accessible.

eDRX cycles can last from seconds to hours (depending on the network operator and the application). The higher the eDRX frequency, the higher the quiescent current consumption of the device. It is therefore crucial to carefully define the required polling frequencies and always set these in relation to the intended service life of the device.

6. ASSEMBLY AND INSTALLATION

6.1. WARNINGS AND SAFETY INSTRUCTIONS FOR MOUNTING

If the sensor is easily accessible after installation, install the sensor first and activate it after installation!

If the sensor is no longer accessible after installation, activate the sensor first and install it after activation!

Before using this type of installation, make sure that the surface on which the sensor is to be screwed is flat, otherwise the housing may be damaged.

Please note:

- Do not insert any objects or body parts into the sensor openings.
- Do not mount the sensor on the ceiling or floor.
- Do not install the sensor at heights of more than two meters.
- Only mount the sensor indoors on a wall in a standard room at a height of 1.50 m to 1.80 m.

The standard versions of the Apollon sensor are designed for a wide range of industrial and logistical applications and have robust housings with a high degree of protection. The following points must be observed during installation:

- Select a mounting location that is within the specified ambient temperatures and conditions (see technical data).
- Do not cover the housing: Radio communication (e.g. LoRaWAN, BLE) must not be obstructed by metallic objects, sealed housings or structure-shielding materials.

Ideally, mount the sensor with a clear line of sight to the sky. This will ensure smooth operation of the radio interfaces.

- Mount the sensor securely, ideally using the mounting holes provided. Low-vibration or solid surfaces are recommended.
- Alignment: The standard version does not require any special alignment and can therefore be mounted flat, vertically or horizontally - depending on the application.
- Do not mount in the immediate vicinity of strong electromagnetic sources to avoid signal interference.

Permanent magnets can generate strong magnetic fields, which can be dangerous if handled incorrectly. Therefore, observe the following warnings:

- - Protect hands and fingers: Strong magnets can suddenly attract and pinch fingers or skin. This can lead to painful crushing and injuries. Always keep sufficient distance when handling permanent magnets and wear protective gloves if necessary.

- - Keep electronic devices away: Magnetic fields can damage electronic devices such as computers, smartphones, credit cards, pacemakers and other sensitive electronics or impair their function. Therefore, always keep a sufficient distance from such devices.
- - Be aware of the risk of breakage: Many permanent magnets are made of brittle materials (e.g. neodymium), which can break in the event of sudden impacts or high loads. The splinters can be sharp and cause injuries. Therefore, use the magnets carefully and avoid impacts or excessive loads.
- - Health risks: People with pacemakers or other implanted medical devices should avoid contact with strong magnets, as the magnetic fields may interfere with or disable these devices. If necessary, consult a doctor before use.
- - Store magnets safely: Keep magnets at a safe distance from each other and from other metallic objects. Sudden attraction can lead to damage, injuries or uncontrollable flying around of the objects.
- Danger for children: Permanent magnets are not toys! Small magnets in particular can be life-threatening if swallowed or inhaled and can cause serious internal injuries. Therefore, always keep magnets out of the reach of children.
- Avoid heating: Permanent magnets permanently lose their magnetic force at temperatures above their maximum operating temperature (between 80 and 200 °C, depending on the material). Therefore, do not expose magnets to direct heat or open flames.

6.2. RECOMMENDED FASTENING METHODS

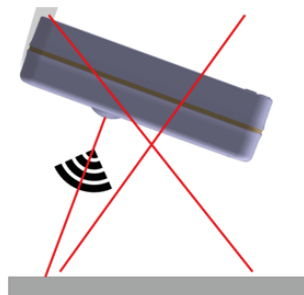
mounting type	description	recommended accessories
screw connection	2 M4 or M5 screws	2x suitable countersunk head screw, wood screw 4mm - 5mm if necessary
magnets	2 neodymium pot magnets M4, internal thread	2x neodymium magnets (indoor) together 16-32 kg load capacity
gluing	Double-sided adhesive tape or mounting adhesive	Double-sided adhesive tape or mounting adhesive

6.3. MOUNTING ACCESSORIES

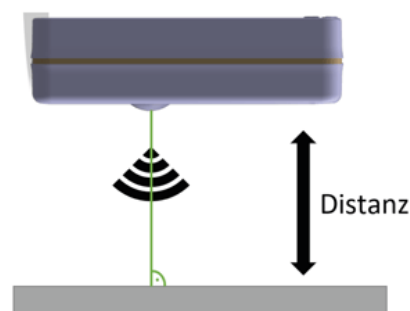
Mounting type	Recommended accessories
screw connection	Machine screws DIN912 with M4 nuts or wood screws
magnets	2 neodymium pot magnets, adhesive force 16 - 32 kg incl. 2 screws
gluing	Double-sided adhesive tape
rivets	For rear wall mounting with M4 rivet nuts or front wall mounting with blind rivets
spacer plate	for dimensions see technical drawing
drilling templates	see end of operating instructions

6.4. GENERAL INSTALLATION INSTRUCTIONS

The sensor measures the distance to the object optimally when it is aligned parallel to it. In the case of lumpy or pourable material, the sensor should be mounted parallel to the bottom of the container. Ideally, the sensor beam hits the object to be measured at an angle of 90°.



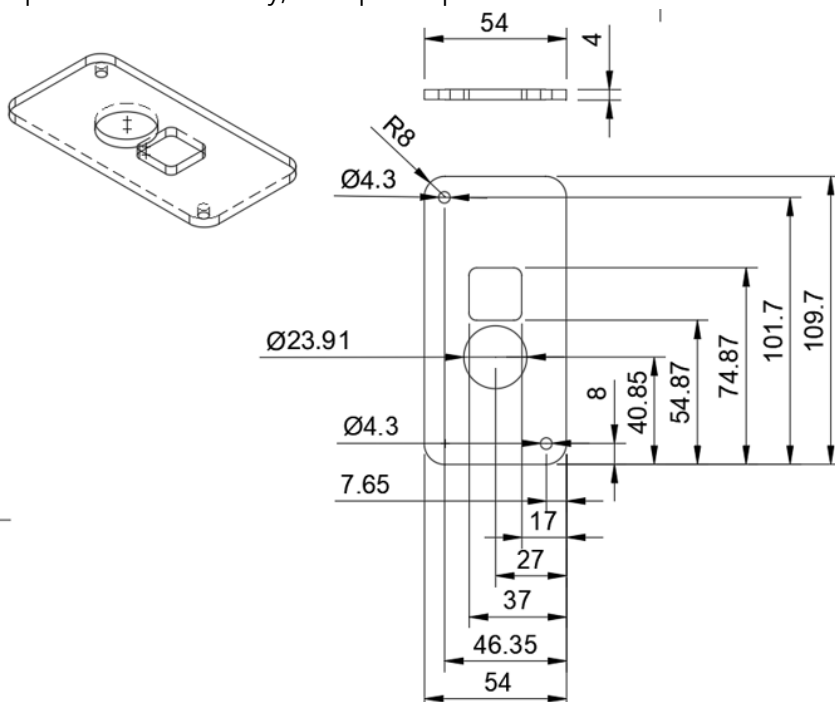
If the sensor is not mounted parallel to the object to be measured, the measuring performance may be impaired. If the sensor is not aligned horizontally, make sure that the percentage calculation of the fill level is adjusted.



6.5. IMPORTANT NOTE FOR DEVICES WITH AN EXTERNAL ANTENNA

If you have ordered a device with an external antenna, which can be recognized by the gold RP-SMA connector, first install the antenna included in the scope of delivery. The sensor should be mounted with the long side on a container. This means that the long side of the sensor and the long side of the container should run parallel. Make sure that the sensor has as free a field of view as possible. The opening angle can be described with a cone of 40°.

Spacer plate: The spacer plate can also be used for front or rear wall mounting. The spacer plate ensures that a flat installation is possible and guarantees the minimum distance between the sensor and the container. In this way, the antenna performance is not impaired. If necessary, the spacer plate can also be used for rear wall mounting.



External antenna: Please note that the antenna should always be mounted vertically and that the tip should point to the sky if the application permits. The antenna should be at least 2 cm away from metal surfaces. Ensure that the antenna is not shielded by surrounding metal parts, if the application allows this.

Internal antenna: If your device has an internal antenna (no external antenna visible), the sensor should always be mounted with the long side vertical, as this allows the maximum signal strength of the device to be achieved. The antenna is located on the top (logo side) of the housing and should be at least 2 cm away from metal surfaces. Ensure that the antenna is not shielded by surrounding metal parts, as far as the application allows.

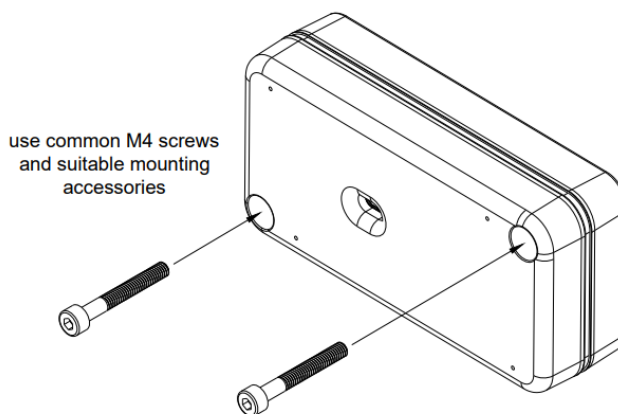
Please note: If you receive a device with an external antenna, never operate the device without an external antenna! This can lead to irreparable damage to the sensor.

6.6. WALL MOUNTING WITH SCREWS

A drilling template can be found at the end of the operating instructions. This type of mounting is generally used to mount the sensor in a container. Insert the appropriate screws into the holes provided on the front. Alternatively, suitable wood or plastic screws can also be used if the sensor is to be attached to the respective materials. Use the two-point bracket supplied for this purpose.

If the sensor is not mounted parallel to the object to be measured, the measuring accuracy may be impaired. If the sensor is not aligned horizontally, the calculation of the level must be adjusted accordingly.

If the sensor is installed on the front wall, a spacer plate should always be used to ensure the necessary distance between the antenna and the wall.



6.7. WALL MOUNTING WITH MAGNETS

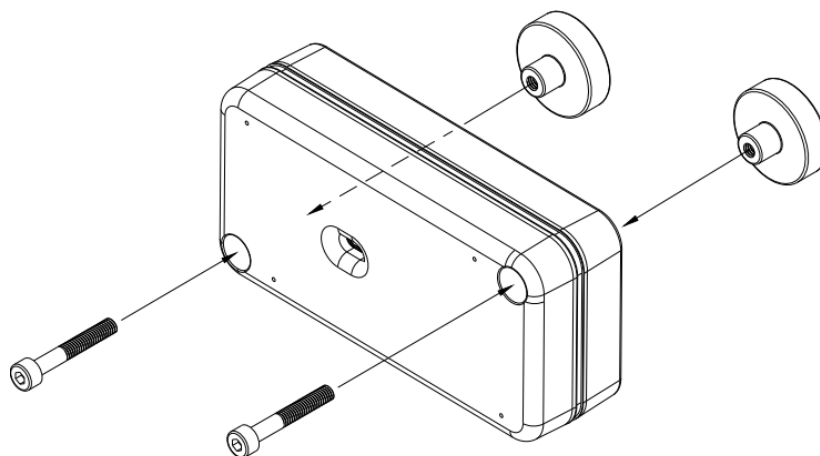
You will find a drilling template at the end of the operating instructions. The magnets shown below are neodymium magnets. Other versions may vary. Although NEODYM magnets offer a higher force per surface area, they are only suitable for outdoor use to a limited extent. Please contact us for advice if required.

Insert the appropriate magnets into the holes provided on the back of the sensor. The magnets shown here are NEODYM magnets. Alternatively, you can carry out this step in the same way for FERRIT magnets.

Insert the screws into the holes on the opposite side and tighten the magnets by screwing them in.

Optional: Pull an anti-slip cover over the magnets. The anti-slip cover prevents the magnet from slipping when the object vibrates or changes position.

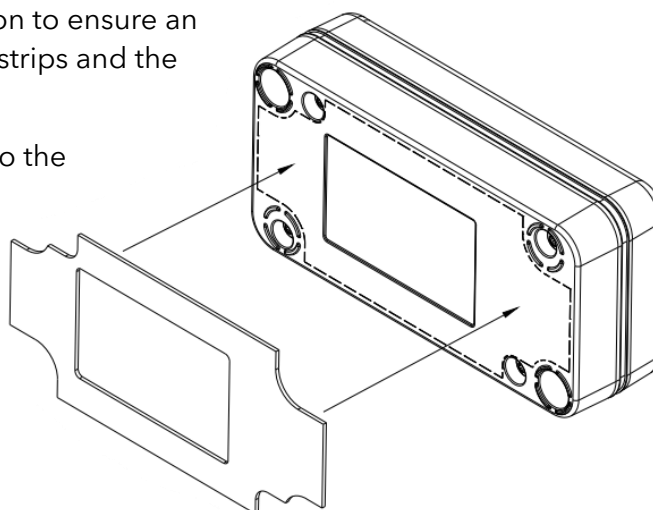
In addition to the rear wall mounting shown, front wall mounting is also possible. To do this, the positions of the magnets are swapped (magnets in the holes on the front, screws in the holes on the back).



6.8. WALL MOUNTING WITH ADHESIVE STRIPS

Make sure that the surface is clean, dry, smooth and adhesive before using this type of installation, as an uneven or dusty surface can impair the adhesion of the adhesive strips. Avoid mounting on rough, porous or damp surfaces, as this can reduce the adhesive strength and cause the mount to peel off. Press the mount firmly for a few seconds after application to ensure an optimum bond between the adhesive strips and the surface.

Stick the double-sided adhesive tape to the back of the sensor in the area shown. The tape should be sufficiently dimensioned to ensure a secure hold. When mounting with adhesive strips, make sure that the adhesive strips adhere completely and that no corners come loose.



6.9. GENERAL INFORMATION ON INSTALLATION WITH BLIND RIVET NUTS

A drilling template can be found at the end of the operating instructions. A blind rivet nut is a type of fastener that is used to insert threads into thin materials such as sheet metal, plastic or wood. It consists of a sleeve with a thread on the inside and a rivet on the outside. The rivet is inserted into the material with a special tool and presses the sleeve together, embedding the thread in the material. The blind rivet nut is often used when it is

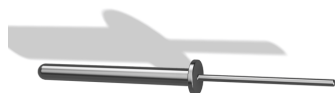
not possible to fit a nut on the back of the material, as the rivet nut is inserted from one side of the material and the thread on the other side is available for screwing in a screw or bolt. A special tool, known as a blind riveting tool, is required for installation.



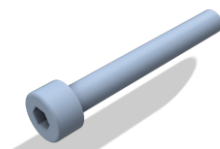
Blind rivet nut
(variant with
countersunk
head)



Spacer sleeve for blind rivets for
insertion into the back of the
housing for front panel mounting



4x40 blind rivets



DIN912 cylinder head screw
M4x40 for front wall mounting
with blind rivet nuts, DIN912
cylinder head screw M4x30 (for
rear wall mounting with blind
rivet nuts)

The following steps are required for **front panel installation with blind rivet nut Fehler!**
Verweisquelle konnte nicht gefunden werden.:

Step 1: Make sure you have all the necessary materials and tools to hand. You will need a blind rivet nut, riveting pliers, a drill, a drill bit, a measuring device and a wrench or pliers.

Step 2: Drill a hole in the material into which the blind rivet nut is to be inserted. The diameter of the hole should be 0.1 mm larger than the outer diameter of the blind rivet nut.

Step 3: Insert the blind rivet nut into the hole by pushing it through the hole.

Step 4: Place the riveting pliers on the blind rivet nut and tighten the pliers to fix the blind rivet nut in place.

Step 5: Use the riveting pliers to insert the blind rivet nut into the material. Tighten the pliers until the blind rivet nut is fully embedded and forms a firm connection.

Step 6: Check the blind rivet nut to make sure it is installed properly and forms a tight connection. If necessary, tighten it with a wrench or pliers to make sure it is securely fastened.

The following steps are required for **rear panel installation with blind rivets Fehler!**
Verweisquelle konnte nicht gefunden werden.:

Step 1: Make sure you have all the necessary materials and tools to hand. You will need a blind rivet, a rivet gun and, if necessary, a drill and a drill bit.

Step 2: If there is no hole yet, drill one at the desired location. The diameter should be about 0.1 mm larger than that of the blind rivet.

Step 3: Inserting the blind rivets Insert the blind rivets into the rivet gun and push them through the hole in the material.

Step 4: Place the rivet gun on the head of the blind rivet and tighten the gun to tension the rivet.

Step 5: Use the rivet gun to set the blind rivet. Tighten the gun until the rivet is fully embedded and forms a secure joint.

Step 6: Check that the rivet is securely in place. If necessary, you can tighten it with a wrench or pliers to ensure optimal fastening.

6.10. FRONT PANEL INSTALLATION WITH RIVET NUTS

Always use the spacer plate for this type of installation to ensure the correct distance between the antenna and the wall.

A drilling template can be found at the end of the operating instruction. Front panel installation is used when you want to mount the **front side** of the Apollon-Q sensor to the wall of the object you are monitoring. The front side is the one with the central opening and visible logo. This mounting method is the **preferred option** because it allows for easy battery or SIM card replacement by simply loosening the screws (unlike front panel mounting using blind rivets).

Make sure to activate the sensor before starting the installation.

Step 1:

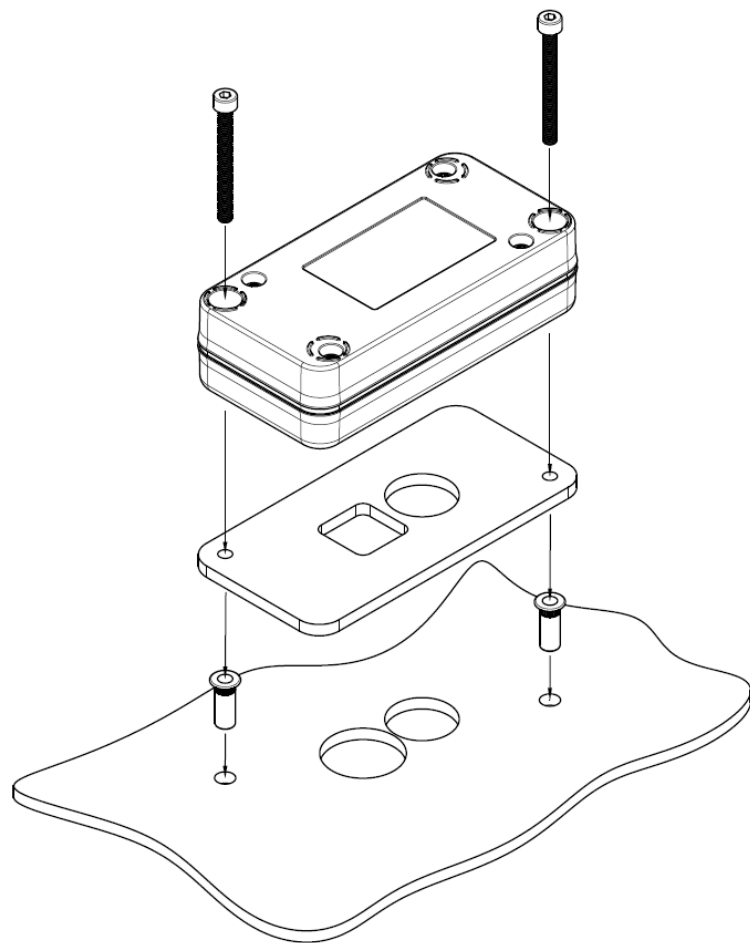
Depending on the sensor type (Radar and ToF only), drill the appropriate holes into the wall where the sensor will be mounted. Use the provided drilling template for this.

The holes for the rivet nuts should have a diameter of 6.1 mm.

Step 2:

Install the rivet nuts using a suitable tool. Use the spacer plate and insert the screws into the mounting tabs as shown in the picture.

Tighten the screws to a maximum torque of 2 Nm. If the sensor will be used in an environment with constant vibration, we recommend applying threadlocker to secure the screws.



6.11. REAR PANEL INSTALLATION WITH BLIND RIVETS

A drilling template is provided at the end of the user manual. Rear panel installation works similarly to front panel installation with rivet nuts.

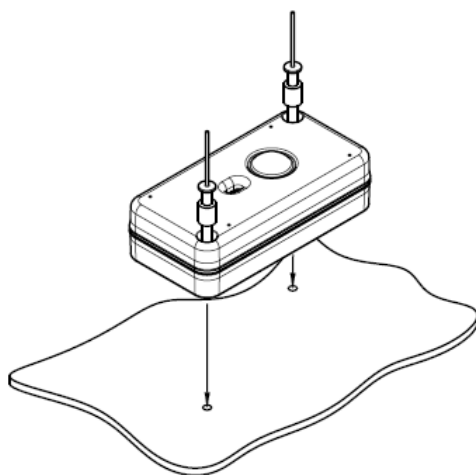
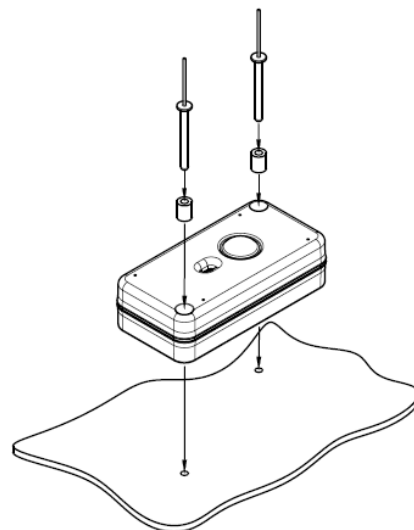
In this case, the **back side** of the sensor is mounted against the wall.

This mounting method uses M4x30 screws and 4x40 blind rivets. The installation is performed without a spacer plate. The clamping range is 3 mm to 6 mm.

The holes for the blind rivets should have a diameter of 4.1 mm.

Step 1:

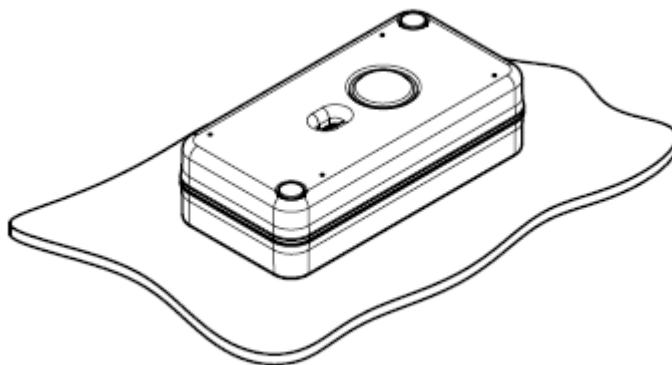
Depending on the sensor type (Radar and ToF only), drill the appropriate holes into the wall where the sensor will be mounted. Use the provided drilling template. The holes for the blind rivets must be 4.1 mm in diameter.



Step 2:

Insert the rivets through the designated mounting tabs. Use the supplied spacers (long version) for this purpose.

First, insert the blind rivet through the spacer, then through the mounting tab on the sensor. Secure the rivets using a suitable rivet tool.



6.12. FRONT PANEL INSTALLATION WITH BLIND RIVETS

Always use the spacer plate for this type of installation to ensure the correct distance between the antenna and the wall.

A drilling template can be found at the end of the user manual. Front panel installation is used when you want to mount the front side of the Apollon-Q sensor to the wall of the object where the sensor is to be installed. The front side is the one with the central opening, where the logo is also visible.

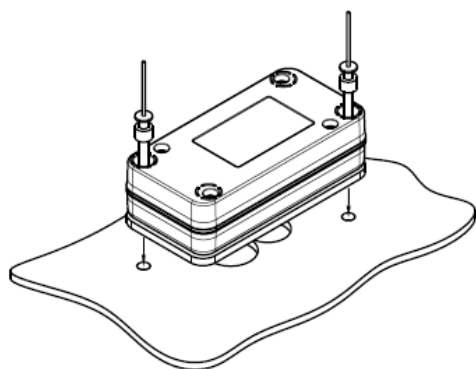
Activate the sensor before starting the installation.

This type of installation uses 4x40 blind rivets. The installation is done using a spacer plate.

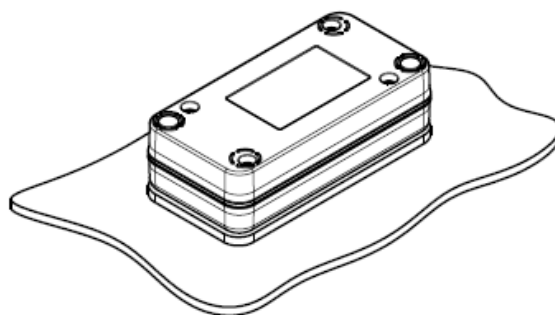
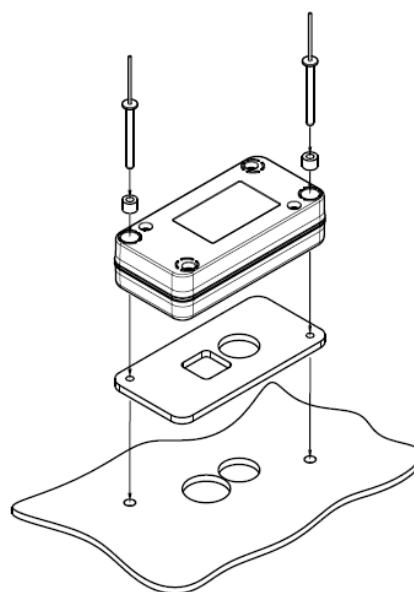
The clamping range is 3 mm to 6 mm.

The holes for the blind rivets should have a diameter of 4.1 mm.

Step 1: Depending on the sensor type (Radar and ToF only), drill the appropriate holes into the wall where the sensor is to be installed. Use the drilling template for this. The holes for the blind rivets should have a diameter of 4.1 mm.



Step 2: Insert the rivets through the designated mounting tabs. Use the included spacers (short version) for this purpose. First insert the blind rivet through the spacer, then through the tabs on the sensor, and finally through the spacer plate. Then secure the blind rivets using a suitable riveting tool.



7. COMMISSIONING AND USE

Please note that using knives or other sharp objects may damage the housing or internal electronics.

There are two Hall sensors (magnetic field switches) located on the sensor. The following diagram shows the position of the Hall sensors and the recommended placement of the magnets.



The upper, center-positioned magnetic switch (1) can be operated in three different modes:

- The container is considered closed when the magnet is in place
- The container is considered open when the magnet is applied
- The sensor registers an opening when the magnet passes over the switch twice

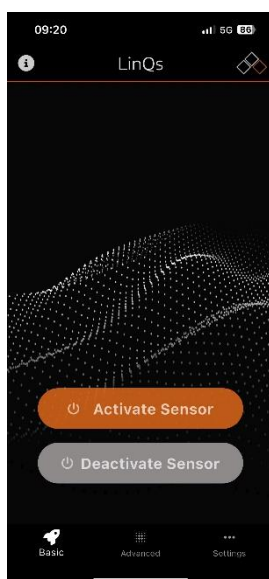
For newer sensor versions, the opening detection function is handled by the accelerometer.

7.1. COMMISSIONING THE SENSOR VIA BLE

BLE is not available for sensors with ToF and without radar. For these variants, please follow the instructions in section 7.3 to commission and configure them.

For all other sensor types, BLE commissioning via smartphone is available:

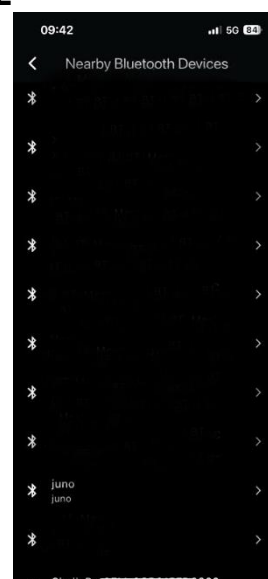
1. All supported variants can be activated via BLE using a smartphone. Simply use our LinQs app, available in the [APP Store](#) and [Google Play Store](#).
2. Activate the sensor by briefly holding a magnet to the lower left Hall sensor (2) for at least 2 seconds. The sensor should respond with an audible signal. Advertising mode will then start automatically.
3. You can now connect to the sensor via BLE from your device. Ensure that BLE is enabled on your smartphone and that you are within range of the sensor to establish the connection.

Activate BLE Advertising

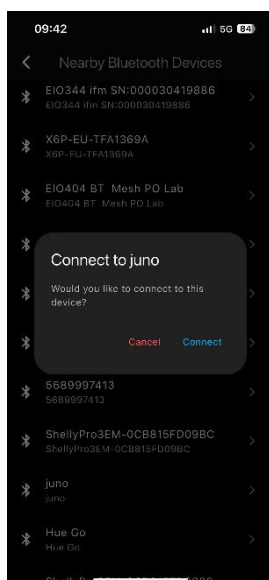
The BLE advertising mode can be activated with the magnet or after activating the sensor.

Search for BLE device

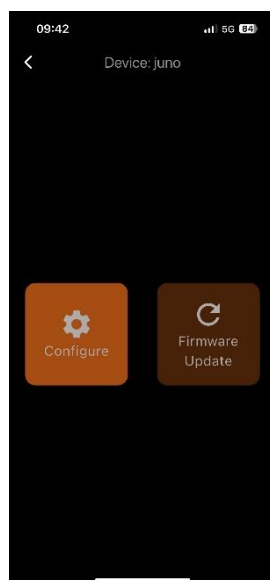
Use the "Search" button to search for the sensor via BLE.

Connect to the sensor via BLE

Select the correct sensor and confirm by clicking on "Apollon".

Connect to the sensor via BLE

Click on the "Connect" button

Configure with the sensor via BLE

Now use the "Configure" button to set parameters.

Configure with the sensor via BLE

A transmission can be triggered using the "Trigger Send" button. Tap the desired table entry and change the values. Confirm with the button below "Update & Reboot" or "Update". Update and Reboot forces a reboot in

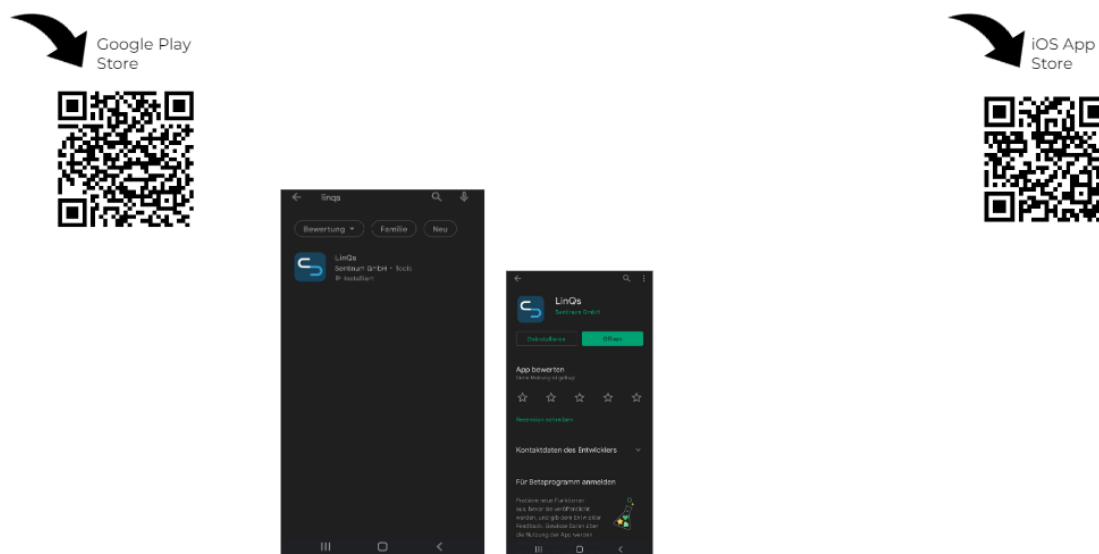
addition to the change,
Update is used for the next
measurement or
transmission.

7.2. AUDIBLE SIGNAL AND FEEDBACK

- When the device is switched on, an acoustic signal consisting of several ascending tones sounds. This sequence of tones signals that the sensor has been successfully activated.
- When the device is switched off, several descending tones are played to acoustically confirm that the device has shut down.
- When establishing or disconnecting a Bluetooth connection (BLE), the sensor also emits an acoustic signal to confirm the connection status.

7.3. NFC COMMISSIONING, PARAMETER CONFIGURATION AND NFC TAG LOCATION

Activation is performed via an NFC app using a smartphone. The app can be downloaded from the respective app stores. Simply search for "Sentinum LinQs" and install the LinQs app.



First, locate the NFC tag on the sensor and the NFC reader on your smartphone. The position of the NFC tag is indicated by the orange arrow on the sensor.



Open the app and activate the sensor. To start the sensor with default settings, tap the "Activate sensor" button on the app's home screen. Then hold your device against the NFC marker on the sensor.



Once the sensor is activated, the message "Sensor updated!" will appear. You can then proceed with activating other sensors.

7.4. POSSIBLE APPLICATIONS OF THE SENSORS

The sensors of the Apollon-Q series are highly versatile and offer reliable solutions for a wide range of fill level monitoring applications. Thanks to the combination of radar and optical measurement, they enable precise measurements across various use cases, whether it involves liquids, bulk materials, or discrete items.

Due to their robust design and support for common communication standards such as mioty®, NB-IoT, LoRaWAN®, and LTE-CAT-M1, these sensors are flexible and ensure reliable data transmission even in demanding environments.

One key area of application is smart waste management. The sensors accurately measure fill levels in trash bins, glass or textile recycling containers, and other waste containers commonly used in cities and municipalities. Precise measurement allows for more efficient collection planning, reduces unnecessary trips, and optimizes waste management logistics. Energy-efficient communication via LoRaWAN® or mioty® enables years of operation without frequent battery replacements.

The sensors also play an important role in industrial and logistics applications. They support continuous monitoring of fill levels in IBC tanks, industrial waste containers, and inventory storage for small parts. With NB-IoT and LTE-CAT-M1 support, measurement data can be transmitted directly via mobile networks, allowing seamless integration into existing IoT systems. This helps improve production workflows and ensures timely restocking.

Another important use case is monitoring in shafts and channels. The sensors measure the fill level of water, sludge, or other materials, enabling better planning of maintenance intervals and early detection of critical conditions. This helps prevent flooding or blockages in time.

Additionally, the integrated GPS function allows tracking of containers and mobile units. This is especially useful in logistics, where both the location and fill level can be monitored at any time to optimize operations.

The Apollon-PP sensor offers reliable presence detection and is ideal for use in logistics and industry. It enables monitoring of shelves, small parts, containers and supports various communication standards such as LoRaWAN®, mioty®, NB-IoT, and LTE-CAT-M1, ensuring easy integration into IoT systems.

Additional features such as opening and vandalism detection, as well as simple provisioning via NFC and downlinks, enhance usability. The sustainable, energy-efficient data transmission and wide range of applications make these sensors a strong solution for modern infrastructure needs.

8. FLAP OPENING DETECTION AND TILT DETECTION

Flap opening detection can be performed either via the magnetic switch or the accelerometer. Tilt detection is handled by the accelerometer.

8.1. FLAP OPENING DETECTION VIA MAGNETIC SWITCH



In older versions, magnetic switch 2 is used for activation. The flap opening is detected via Hall switch 1. The magnetic switches can be operated in three different modes:

1. The magnetic switch is active. One or both sensors can be used.
2. Large neodymium magnets are recommended. These should be placed as close to the sensor as possible. Due to the varying size of magnets, a universal distance cannot be specified. However, a maximum distance of 1 cm between the magnet and the sensor housing is recommended.
3. For comparison: A neodymium disc magnet with a diameter of 20 mm and height of 5 mm achieves reliable detection at distances under 1 cm.
4. The magnetic switches support the following three operating modes:
 - The container is closed when the magnet is present
 - The container is open when the magnet is applied
 - The sensor registers an opening when the magnet passes by twice

8.2. FLAP OPENING DETECTION VIA ACCELEROMETER AND TILT DETECTION

In newer versions, only magnetic switch 2 is used to activate BLE advertising.

The Apollon sensor is equipped with a built-in 3-axis accelerometer that reliably detects changes in position and motion. One of its key features is the detection of flap, lid, or housing openings, which are common in industrial applications.

1. Position detection at rest:
 - When the flap is closed, the sensor remains in a defined, stable position
 - The LIS2DTW12 continuously measures acceleration along the X, Y, and Z axes
 - Based on static acceleration (mainly due to gravity), the absolute position of the flap can be determined
2. Tilt or movement detection:
 - When the flap is opened or moved, the sensor's spatial orientation changes
 - This change is detected by significant deviations in acceleration on at least one axis
 - The deviation is interpreted as a trigger event
3. Threshold-based detection:
 - The Apollon firmware allows you to define a tilt or motion threshold (e.g., a change of 15°, not ultra-low-power mode)
 - Once this threshold is exceeded, an opening event is recorded

The sensor can still operate in an energy-efficient mode by adjusting the tilt measurement interval to a longer period, such as every 5 minutes. In such cases, power consumption is negligible.

4. Optional: Interrupt-driven operation:
 - The sensor supports low-power modes with interrupt triggering
 - This means it stays in a low-power state and only wakes the microcontroller upon motion detection – ideal for maximizing battery life
 - **Limitation: The tilt angle cannot be configured and is fixed at 65°**
5. Event handling and data transmission:
 - After a flap opening is detected, the event is logged in internal memory
 - Depending on the configuration, a data packet can be sent immediately via LoRaWAN, BLE, or another network protocol to report the event

Advantages of this method:

- No mechanical components required (unlike reed or magnetic switches)
- Immune to magnetic field interference
- Easily retrofitted or reconfigured via software

8.3. ORIENTATIONS (EXAMPLE APOLLON IQ)

Front view



Side view



Rear view



9. COMMUNICATION WITH THE INTERFACE

Detailed configuration options for sensor communication and join behavior can be found in the respective documentation for [LoRaWAN®](#), [mioty®](#), or [cellular \(NB-IoT and LTE-M1\)](#), depending on the version.

All generic documentation for sensor platforms is available at:

<https://docs.sentinum.de/en/important-cross-product-documentation-for-sensors>

9.1. LORAWAN JOIN BEHAVIOR

Before telemetry data can be transmitted via LoRaWAN, the device must successfully join the network. To achieve this, it sends join requests repeatedly until a join accept is received. As a trade-off between energy consumption and quick network access, the transmission intervals between join requests increase progressively. The device also changes its data rate over time—starting with a high data rate (low spreading factor) and shifting to a lower data rate (high spreading factor).

The join procedure strictly follows the specifications and recommendations of the LoRa Alliance. Sentinum sensors implement this through so-called join bursts, where the interval between bursts grows over time.

Each join burst consists of up to 6 join requests, sent with decreasing data rates (DR5 to DR0) or increasing spreading factors (SF7 to SF12). The intervals between individual requests within a burst grow quadratically to comply with the LoRa Alliance's duty cycle limitations.

The duty cycle constraints are defined as follows:

- <1h: 1%
- <11h: 0.1%

This means the same transmission budget is available during the second phase (<11h) as in the first phase (<1h), even though the time frame is ten times longer. To fully utilize the available budget, join bursts are initially spaced closely and then spaced further apart.

In detail:

- Phase 1: 2 bursts are sent (burst duration ~10 minutes)
- Phase 2: 2 more bursts (~100 minutes)
- Phase 3: 1 burst per day (burst duration up to 16 hours)

9.2. MIOTY JOIN BEHAVIOR

Before telemetry data can be sent via mioty, the device must first establish an initial communication path with the base station. Unlike LoRaWAN, mioty does not require an explicit join procedure. Instead, the device begins immediately transmitting so-called

telegrams. Each telegram consists of 512-bit packets, which are split into up to 12 sub-packets and transmitted over multiple frequencies and time slots (telegram splitting).

The backend interprets the first successfully received telegram (with a valid device ID) as a network join. Device configuration (e.g., Device ID, Application Key) must be preconfigured to match backend settings.

If no confirmation is received by the backend (e.g., due to missing reception timestamps or decoding failures), the device retries transmission. These retries follow mioty send cycles. The cycle duration increases progressively—from 5 minutes (after the first attempt), to 30 minutes, and up to 12 hours—to comply with duty cycle limits (<1%) and minimize energy consumption.

The transmission frequencies (868.0–868.6 MHz in Europe) and channels are rotated to avoid collisions. The data rate is fixed by the protocol:

- 15.625 kbit/s for uplink
- Optionally 4.882 kbit/s for downlink

Telegram transmissions are robust to interference. Only 3 of the 12 sub-packets need to be received correctly thanks to forward error correction.

Each full transmission attempt lasts about 1.2 seconds (including guard time). Once successfully received, the device switches to normal operation with fixed transmission intervals. If no reception occurs after several attempts (typically 10–12), the device pauses for 24 hours before starting a new initialization cycle.

9.3. CELLULAR JOIN BEHAVIOR (NB-IOT AND LTE-M1)

Before telemetry data can be sent via cellular networks (NB-IoT or LTE-M1), the device must complete the standardized network attachment procedure as defined by 3GPP (Release 13 and later). This involves scanning the LTE frequency bands (e.g., Band 8 or Band 20 in Europe) for available cells. Once synchronization (PSS/SSS) and system information decoding (SIB) are successful, the device initiates the attach process, which includes:

- RRC Connection Request
- NAS Authentication
- Security setup
- Establishment of a PDP or PDU context (APN, IP address)

If the initial attempt fails (e.g., due to no signal, rejected authentication, or missing PDU context), a retry mechanism starts. Retry intervals increase exponentially—starting at around 15 seconds, then 60 seconds, 5 minutes, and up to 6 hours. The maximum number of join attempts per day is often limited by the network carrier (e.g., 6–8 attempts). Cellular modems strictly follow 3GPP and GSMA guidelines on access control.

Both NB-IoT and LTE-M1 support Coverage Enhancement Levels (CE Levels 0–2), enabling connection even in poor signal conditions. In CE Level 2, a message can be repeated up to 2048 times.

Typical data rates:

- NB-IoT: up to 250 kbit/s uplink, 26 kbit/s downlink
- LTE-M1: up to 1 Mbit/s uplink

Connection time depends on network conditions and can take several hours in worst-case scenarios (e.g., CE Level 2 with backoff).

To reduce energy consumption, cellular devices use the following power-saving mechanisms after successful join:

- PSM (Power Saving Mode): The device remains in low-power sleep mode for hours or days and reconnects only at the next scheduled transmission.
- eDRX (Extended Discontinuous Reception): The device checks for downlink messages only at predefined intervals (e.g., every 20 minutes to 3 hours).

If no network is found, the device aborts its search after a maximum of 60 minutes and enters deep sleep for several hours. It then starts a new search and join cycle. This behavior is optimized by the manufacturer according to carrier recommendations and the need for energy efficiency in large-scale IoT deployments.

10. MAINTENANCE AND CLEANING

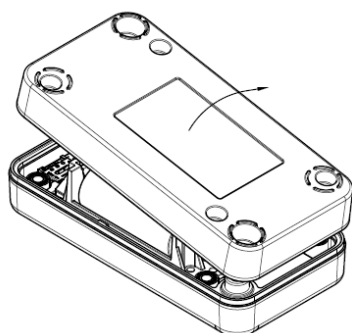
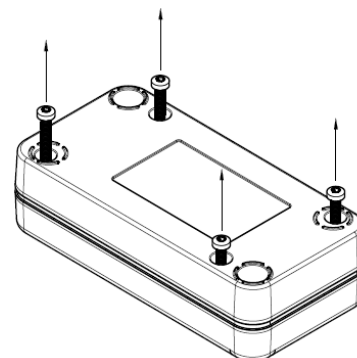
To ensure reliable sensor operation and long service life, regular maintenance is recommended. Please observe the following guidelines:

- Clean the housing—especially the sensor's ventilation slots—using a dry or slightly damp microfiber cloth. Ensure that no moisture enters the device.
- Perform cleaning regularly, especially in dusty or pollen-rich environments, to maintain the long-term functionality of the sensor.
- Do not use alcohol-based or solvent-based cleaning agents, as these may damage the sensor's surface.
- Avoid using compressed air or other intensive cleaning methods, as they may harm sensitive sensor components.
- Hard deposits (e.g., lime, oil, or grease) can impair measurement accuracy. If necessary, remove them early using a soft, damp cloth and a mild cleaning agent.
- For optical or radar-based sensors, regularly inspect lenses or antenna surfaces for dirt or scratches.
- Periodically check that the sensor and its mounting are securely in place.

11. BATTERIE CHANGE

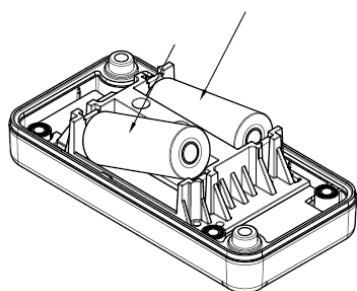
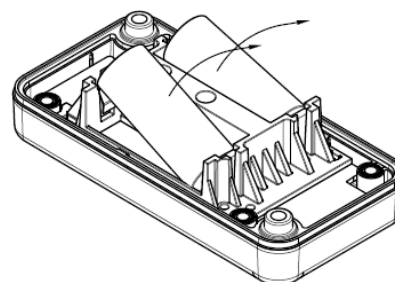
communication standart	approved batteries
LoRaWAN® and mioty®	• Energizer® Ultimate Lithium-Batterien - AA • VARTA Ultra Lithium Mignon AA
cellular sensors (NB-IoT and LTE-CAT-M1)	• VARTA-CR-AH-R A

Remove the four screws on the back of the sensor indicated by the orange arrows. Use a Torx T10 screwdriver. Ensure the sealing gasket is undamaged during disassembly.



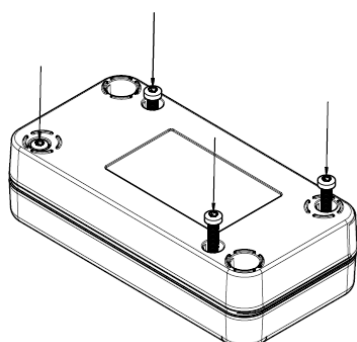
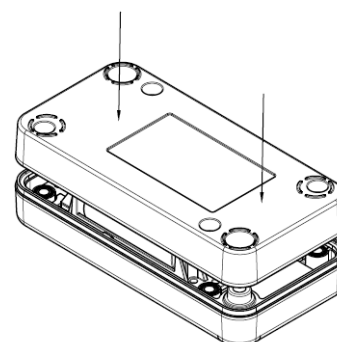
Carefully lift off the back cover of the sensor enclosure.

Remove the used battery cells out of the battery holder.



Insert two new AA-sized battery cells. Using batteries other than the ones listed above may affect performance and product safety. Specified runtimes and functionality may not be guaranteed with non-approved battery types.

After inserting the batteries, the sensor should emit a short beep indicating successful startup. Place the back cover back onto the enclosure.



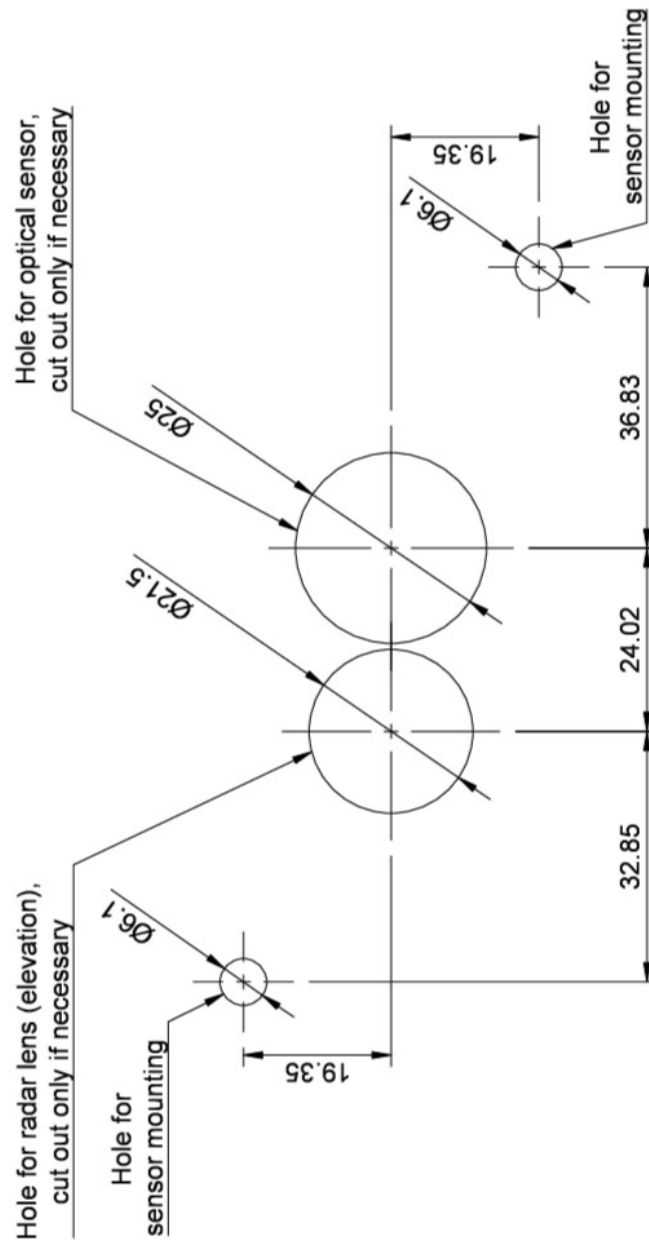
Tighten the screws in a crisscross pattern to ensure even pressure distribution and avoid mechanical stress. Check that the gasket remains in its original position without any deformation. Mount the sensor back in its operating position. Dispose of the used batteries in accordance with local environmental regulations.

12. MARKING AND CERTIFICATION

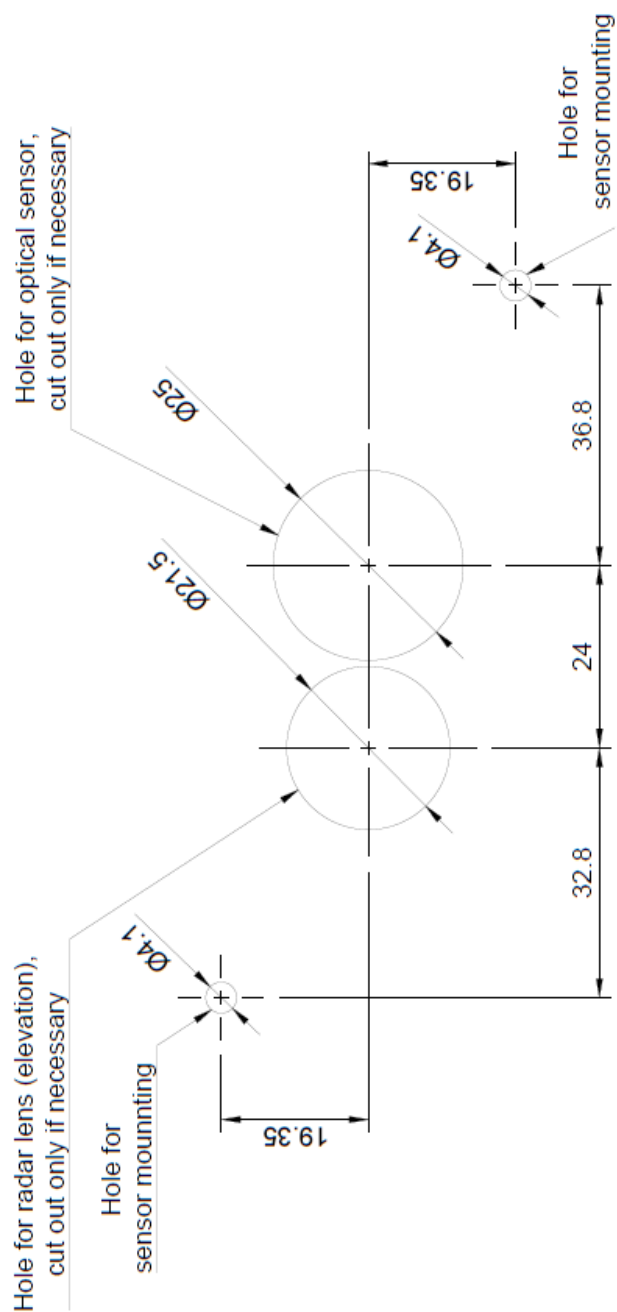


13. DRILL TEMPLATES FOR INSTALLATION

13.1. BLIND RIVET NUT MOUNTING



13.2. BLIND RIVET MOUNTING



13.3. WALL MOUNTING (SEE NEXT PAGE)

